PS1

Controlled source

↕

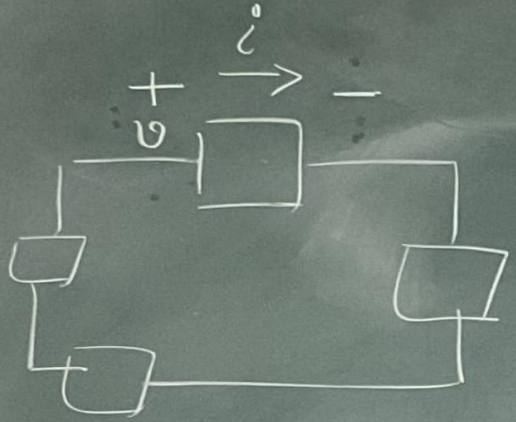
Dependent source

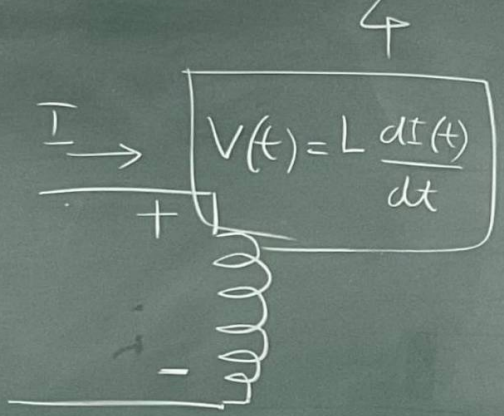
VCVS

↑ Volt ↑ Volt

$v, \mu \in \mathbb{R}$







VCCS

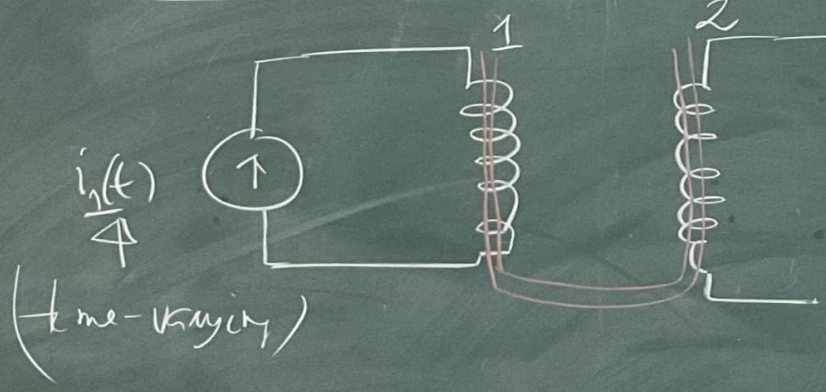
↑ Curr

CCVS

CCCS

PS2

Mutual Inductance



$\phi_2(t) \propto i_1(t)$

(±) Compl

$v_2(t) \propto \frac{di_1(t)}{dt}$

M_{21}

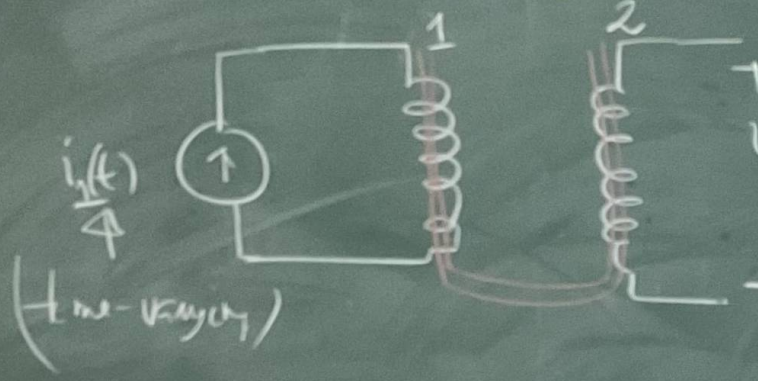
(+)

CCVS

CCCS

P52

Mutual Inductance

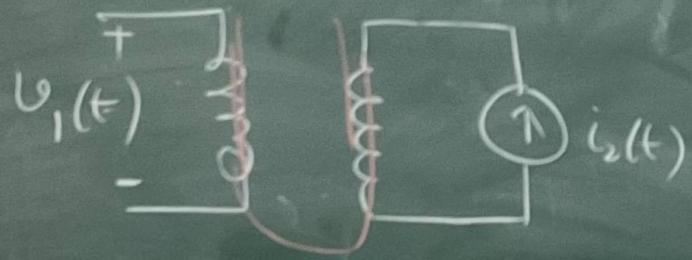


(+ mutual coupling)

$\phi_2(t) \propto i_1(t)$

$v_2(t) \propto \pm \frac{di_1(t)}{dt}$ (±) Coupling

M_{21}



$v_1(t) \propto \pm \frac{di_2(t)}{dt}$ (±) (symmetry) Energy Cons.

M_{12}

$M_{21} = M_{12} = M$

P54

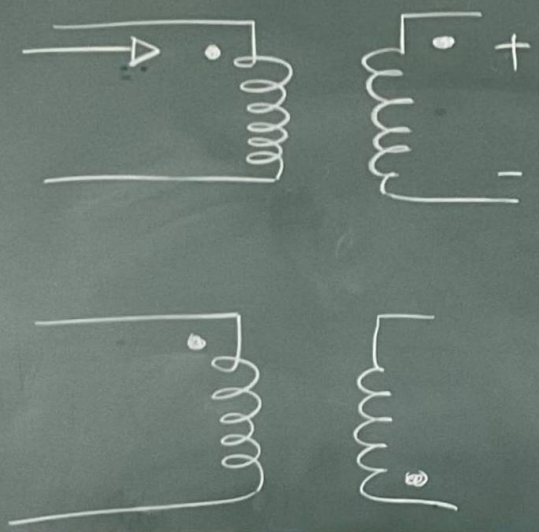
i_1

PS3

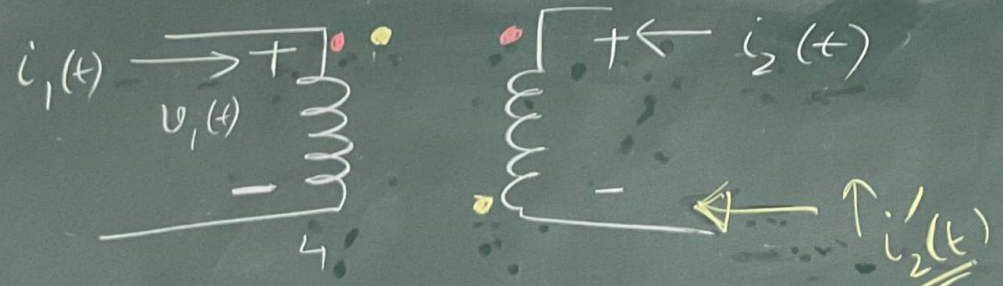
Engt to decide sign & codify it

Dot convention

A current (incr ↑) entering dot side of a coil will create +ve voltage on the dot side of the other coil.



PS4



$v_2(t) = L_2 \frac{di_2(t)}{dt}$ if $i_1(t) = 0$

$v_1(t) = L_1 \frac{di_1(t)}{dt}$ if $i_2(t) = 0$

$v_1(t) = L_1 \frac{di_1(t)}{dt} + M \frac{di_2'(t)}{dt}$

(PS 4)



$v_2(t) = L_2 \frac{di_2(t)}{dt}$ if $i_1(t) = 0$

$v_1(t) = L_1 \frac{di_1(t)}{dt}$ if $i_2(t) = 0$

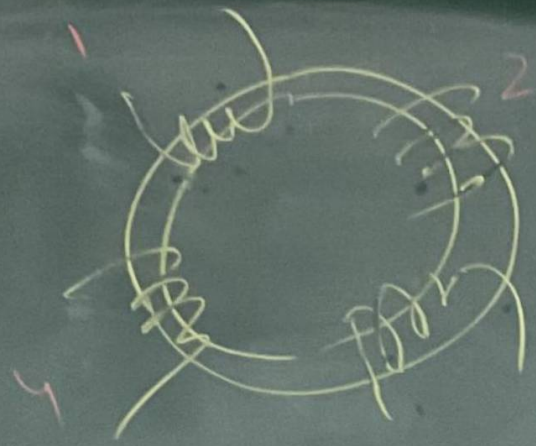
$$v_1(t) = L_1 \frac{di_1(t)}{dt} + M \frac{di_2(t)}{dt}$$

$$v_2(t) = L_2 \frac{di_2(t)}{dt} + M \frac{di_1(t)}{dt}$$

$$v_1(t) = L_1 \frac{di_1(t)}{dt} + M \frac{di_2(t)}{dt}$$

$$- M \frac{di_2(t)}{dt}$$

$$v_2(t) = L_2 \frac{di_2(t)}{dt} - M \frac{di_1(t)}{dt}$$



$L_1 \quad L_2 \quad L_n$

$$\begin{bmatrix} L_1 & M_{12} & M_{13} & \cdots & M_{1n} \\ M_{21} & L_2 & & & \\ \vdots & & & & \\ M_{n1} & & & & L_n \end{bmatrix}$$

(sym / energy conv)

$M_{ij} = M_{ji}$

PDF

$$\begin{bmatrix} v_1(t) \\ \vdots \\ v_n(t) \end{bmatrix} = e^{[A]t} \begin{bmatrix} v_1(0) \\ \vdots \\ \end{bmatrix}$$

$$\begin{bmatrix} v_1(t) \\ \vdots \\ v_n(t) \end{bmatrix} = [A] \begin{bmatrix} \frac{di_1(t)}{dt} \\ \frac{di_2(t)}{dt} \\ \vdots \end{bmatrix}$$

NOT needed for this course

P26

$$v_1(t) = L_1 \frac{di_1(t)}{dt} + M_{12} \frac{di_2(t)}{dt}$$

signed

$$v_2(t) = L_2 \frac{di_2(t)}{dt} + M_{21} \frac{di_1(t)}{dt}$$

Laplace

$$V_1(s) = L_1\left(sI_1(s) - i_1(0)\right) + M_{12}\left(sI_2(s) - i_2(0)\right)$$

P26

$$v_1(t) = L_1 \frac{di_1(t)}{dt} + M_{12} \frac{di_2(t)}{dt}$$

Signed

$$v_2(t) = L_2 \frac{di_2(t)}{dt} + M_{21} \frac{di_1(t)}{dt}$$

Laplace

$$V_1(s) = L_1\left(sI_1(s) - i_1(0)\right) + M_{12}\left(sI_2(s) - i_2(0)\right)$$

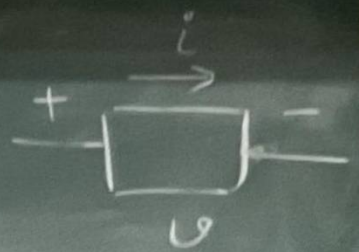
$$V_2(s) = \text{W}$$

$$\begin{bmatrix} V_1(s) \\ V_2(s) \end{bmatrix} = s \begin{bmatrix} L_1 & M_{12} \\ M_{21} & L_2 \end{bmatrix} \begin{bmatrix} I_1(s) \\ I_2(s) \end{bmatrix} - \begin{bmatrix} L_1 & M_{12} \\ M_{21} & L_2 \end{bmatrix} \begin{bmatrix} i_1(0) \\ i_2(0) \end{bmatrix}$$

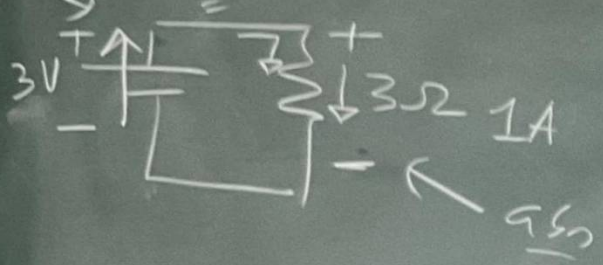
PJ 7

Absorbing power/energy $\Leftrightarrow U \cdot i > 0$

Releasing " $\Leftrightarrow U \cdot i < 0$



Releasing



Resistor voltage drop is always along current

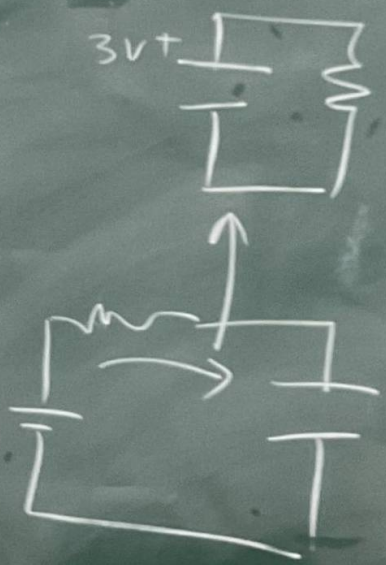
$\Rightarrow$ always absorbing!

dissipative

passive

$$i(t) = C \frac{dv(t)}{dt}$$

$$v(t)\, i(t) = C\, v(t) \frac{dv(t)}{dt}$$



$v(t)$ +ve & ↓ discharging ⇒ releasing energy

+ve & ↑ ⇒ absorbing energy

charging

non-dissipative